Comments from Dr. Consi:

1. Good project
2. Would have liked to see an experiment or two with different Kalman Gains
3. Needs short review of relevant research by others.
4. Grades:
    a. Paper: 95
    b. Presentation:100 (good discussion)

EECE 5554

ROBOTICS SENSING & NAVIGATION

LiDAR-IMU Sensor Fusion for Robot Odometry

Group 5

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1. Problem Statement

The purpose of this project is to implement and evaluate Lidar-IMU sensor fusion for local robot odometry using the KITTI dataset. Accurate estimation of a robot's pose is critical for reliable navigation and motion planning and mapping. Relying on a single sensor introduces significant errors: IMU-only estimation accumulates drift over time, while Lidar-only odometry suffers from motion distortion and scan-matching failures in feature-deprived environments.

This project investigates:

- The standalone performance of IMU-only dead-reckoning odometry and Lidar-only scan-matching odometry.
- The improvement achieved by fusing Lidar and IMU measurements via an Extended Kalman Filter (EKF).
- How long the fused system can operate reliably before global correction becomes necessary
- Failure modes and limitations of the fusion pipeline.

2.Overview

2.1 Dataset

Three sequences in KITTI Odometry/SLAM dataset were used for all experiments.

Sequence 00: Long(3.7kms), urban, medium speed sequence with loop closures. Consists of 4541 frames

Sequence 01: Medium-length, Highway(feature-deprived),high speed sequence. Consists of 1100 frames

Sequence 04: Short, rural, low speed sequence. Consists of 271 frames

Sensor	Model	Key Specification
Lidar	Velodyne HDL-64E	64 beams, 10 Hz scan rate, 100 m range
IMU / GPS	NovAtel SPAN (OxTS)	Tactical-grade, 100 Hz IMU, dual-frequency GPS
Platform	VW Station Wagon	Urban/rural/highway driving scenarios
Ground Truth	GPS/INS Fusion (OxTS)	High-accuracy reference trajectory

Table 1: Hardware used for KITTI data acquisition

2.2 Software Pipeline Overview

The pipeline consists of four major components, implemented across four Python modules:

- **kitti_loader.py**: Loads all sensor data, calibration transforms (IMU-Velo), ground truth poses, and synchronizes IMU readings with Lidar frames.
- **imu_preintegration.py:** Implements IMU preintegration on $SO(3) \times R^3$, propagating rotation, velocity, and position increments between IMU timestamps and Lidar frames as well.
- **lidar_scan_matching.py**: Performs voxel-downsampled ICP (Iterative Closest Point) scan-to-scan matching using Open3D to extract relative pose transforms between consecutive Lidar frames, Parameters : **Number of iterations =20, Voxel Size = 0.5**
- **sensor_fusion_ekf.py**: Implements the EKF that uses IMU preintegration for the predict step and ICP-derived relative transforms for the update (measurement) step and computes the fused trajectory.

2.3 Three Evaluated Configurations

Three odometry configurations were evaluated to isolate the contribution of each sensor:

1. **IMU-Only**: Dead reckoning via IMU preintegration from the first ground truth pose, without any Lidar correction.
2. **Lidar-Only**: Trajectory built by chaining consecutive ICP transforms from an initial known pose.
3. **Lidar-IMU EKF Fusion**: IMU preintegration provides the EKF prediction step; ICP provides the measurement update. The IMU initial relative pose transformation is passed to ICP to improve convergence speed and robustness.

3. Methodology

3.1 IMU Preintegration

The IMU preintegration module accumulates raw IMU measurements (gyroscope and accelerometer) between consecutive Lidar frames k and k+1. Rather than integrating from the world origin each time, incremental preintegrated terms are computed between IMU timestamps:

- $\Delta R \in SO(3)$: Rotation increment, computed using the exponential map (Rodrigues formula).
- $\Delta v \in R^3$: Velocity increment in the IMU body frame.
- $\Delta p \in R^3$: Position increment in the IMU body frame.

Gyroscope and accelerometer biases (bg, ba) are modeled as random walks as the biases are not constant and maintained in the 15-dimensional EKF state. First-order bias Jacobians (J_R_bg, J_v_bg, J_v_ba, J_p_bg, J_p_ba) are propagated alongside the preintegrated measurements, enabling efficient bias correction without full re-integration. A 15×15 process noise covariance is also maintained using state transition and noise input matrices.

Noise parameters used are consistent with automotive-grade IMU specifications: $\sigma g = 1.7\times10^{-4}$ rad/s/√Hz (gyroscope), $\sigma a = 2.0\times10^{-3}$ m/s²/√Hz (accelerometer).

3.2 Lidar Scan Matching (ICP)

For each pair of consecutive LiDAR frames, point cloud preprocessing and registration are performed:

- Voxel downsampling: Point clouds are downsampled using a voxel size of 0.5 m to reduce computational load while preserving geometric structure.
- Normal estimation: Surface normals are computed via KD-tree-based neighborhood search (radius = 1.0 m, max neighbors = 30) to support point-to-plane variants.
- ICP registration: Open3D's point-to-point ICP is applied with a maximum correspondence distance of 1 m (1.5× voxel size).
- IMU predicted relative transform: In the fused pipeline, IMU-predicted relative transform is passed as the initial alignment for ICP.

The output of ICP is a 4×4 rigid transform T_ICP representing the relative motion of the vehicle between consecutive scans. The ICP RMSE is monitored as a quality metric.

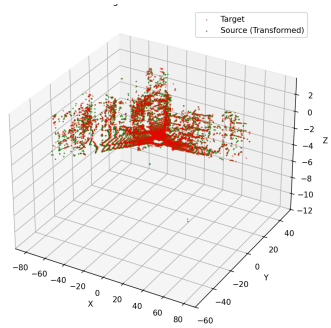


Figure 1: Lidar Point Cloud Matching, using Open3d for sequence 00 of KITTI dataset(Frame 0→ Frame1)

3.3 EKF based Sensor Fusion

The EKF operates on a 15-dimensional error-state vector: [δθ, δv, δp, δbg, δba], representing perturbations in rotation (Lie algebra), velocity, position, gyroscope bias, and accelerometer bias, all in the world frame.

Predict Step

The EKF prediction propagates the nominal state using the preintegrated IMU measurements and gravity. The 15×15 state transition Jacobian F captures how errors in the previous state map to errors in the propagated state, including cross-coupling between rotation error, velocity error, and bias. The preintegration covariance serves as the process noise Q.

State Propagation: $x_{t+1} - g_t(x_t, u_t) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, R_t)$

Preintegration Output: $\hat{\mu}_{t+1} - g_t(\mu_t, u_t) \longrightarrow \Delta R, \Delta v, \Delta p$ from preintegration

Update Step

When a new ICP transform T_ICP is available, a 6-dimensional innovation is computed: the translation residual (difference between ICP-measured relative translation and EKF-predicted relative translation) and the rotation residual (using the SO(3) log map). The 6×15 measurement Jacobian H maps the state perturbations to the measurement space. The Kalman gain K is computed from H, the prior covariance P, and the measurement noise R. The Joseph form covariance update is used for numerical stability. One thing to point here is that the measurement noise is a tunable in this process with sigma_r = 0.01 and sigma_t = 0.05 this values are chosen , to see if IMU-based pose estimation, really improves when we fuse it with the lidar measurements.

Linearise measurement model about $\hat{\mu}_t$:

$$h_t(x) \approx h_t(\hat{\mu}_t) + H_t(x - \mu_t) + \mathcal{O}(||x - \mu_t||^2)$$

where $H_t \widehat{=} \frac{\partial h_t}{\partial x}|_x \in \mathbb{R}^{6\times 15}$:

$$H_t - \begin{bmatrix} I_3 & 0 & 0 & 0 & 0 \\ 0 & 0 & R_k^1 & 0 & 0 \end{bmatrix}$$

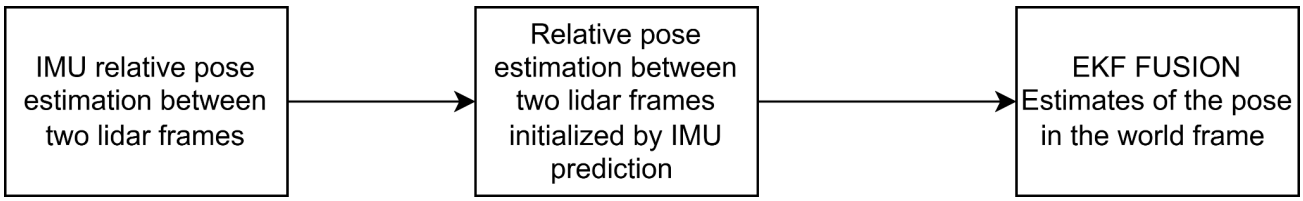
$$\text{Innovation: } y - z_t - h_t(\hat{\mu}_t) - \begin{bmatrix} \log_{SO(3)}(\Delta R_{imu}^T \cdot R_{lidar} \\ t_{lidar} - \Delta p_{imu} \end{bmatrix} \in \mathbb{R}^6$$

$$1.\ \text{Kalman Gain: } K_t - \hat{\Sigma}_t H_t^T (H_t \hat{\Sigma}_t H_t^T + Q_t)^{-1} \in \mathbb{R}^{15\times 6}$$

$$2.\ \text{Posterior Mean: } \mu_t - \hat{\mu}_t + K_t(z_t - h_t(\hat{\mu}_t))$$

$$\text{Inject correction into nominal state: } R \leftarrow R \cdot \exp_{SO(3)}(\delta\phi), v \leftarrow v + \delta v, p \leftarrow p + \delta p$$

$$3.\ \text{Posterior Covarianve (Joseph Form): } \Sigma_t - (I - K_t H_t)\hat{\Sigma}_t (I - K_t H_t)^T + K_t R_{meas} K_t^T$$



3.4 Data Loading and Synchronization

The KittiLoader class handles all calibration transforms (IMU-to-Velodyne and its inverse), loads OxTS GPS/INS data at the raw acquisition rate, and timestamps both Lidar and IMU streams. For each Lidar frame interval [k, k+1], all IMU readings falling within that time window are returned by get_imu_between(), supporting both the KITTI synced (1:1 IMU:Lidar) and raw unsynced (high-rate IMU) configurations. Ground truth poses are derived from OxTS GPS/INS via Mercator projection and Euler-angle rotation matrices, expressed in the world frame.

The preintegrated IMU deltas are re-expressed in the Lidar frame using the extrinsic calibration 'T_imu_to_velo' before the EKF predict step to ensure a consistent measurement frame. These Lidar measurements are then incorporated into EKF based state errors which expresses nominal states in the world frame

4. Evaluation Metrics

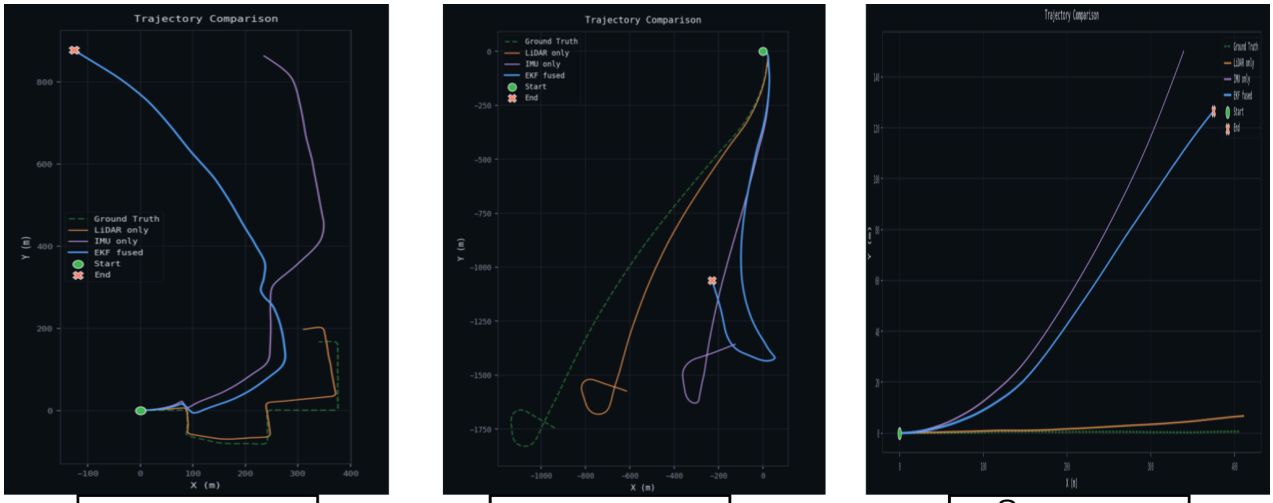
The following quantitative metrics are used to compare the three odometry performances against the OxTS GPS/INS ground truth:

- **Trajectory Comparison** : To analyse how the trajectory drifts
- **Drift (Instantaneous Translation Error) vs Time & Distance**: To assess the drift of IMU and Lidar with distance and time, to understand reliability of each sensor for pose estimation, for safe limits
- **ICP RMSE vs Number of Frames** : To assess ICP convergence in all of the scenes
- **Loop Closure Analysis**: To understand, if the loop closures(global correction) improve the accuracy of the pose estimation.

5. Results and Analysis

5.1 Trajectory Comparison

IMU-trajectory vs Lidar-trajectory vs Fused-trajectory



Sequence 00 Sequence 01 Sequence 04(Baseline)

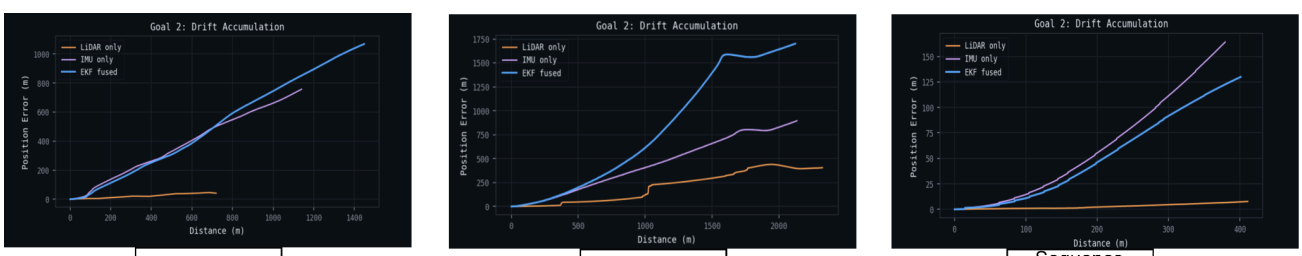
Figure 2: Trajectory Comparison for sequences(00,01,04 in KITTI

5.2 Sensor Drift (Instantaneous Translation Error) vs time

Sequence 00 Sequence 01 Sequence 04(Baseline)

Figure 3: Drift vs Time for sequences(00, 01,04) in KITTI

5.3 Sensor Drift (Instantaneous Translation Error) vs distance



Sequence 00 Sequence 01 Sequence 04(Baseline)

Figure 4:Drift vs Distance for sequences(00, 01,04) in KITTI

5.3 ICP RMSE values for all Sequences:

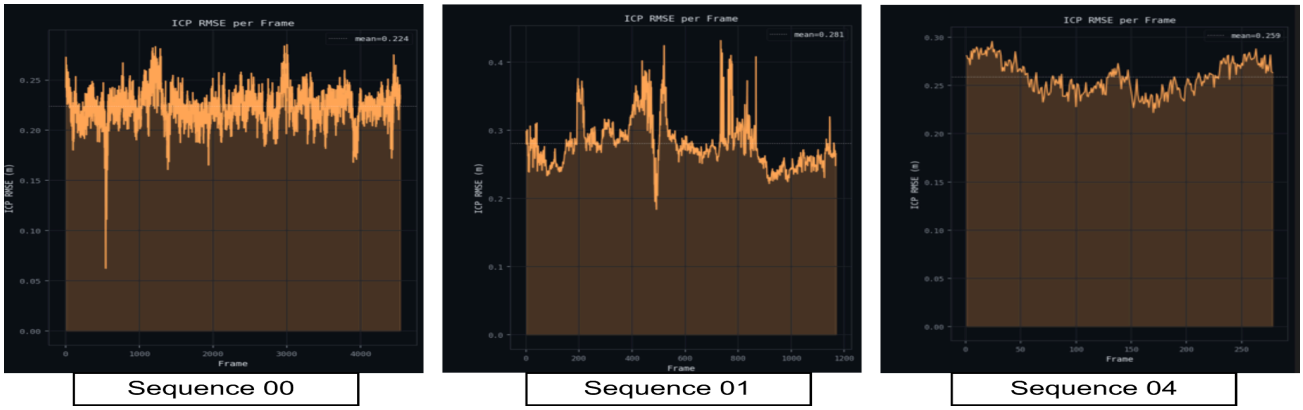


Fig 5: ICP RMSE vs No. of Frames for sequences(00, 01,04) in KITTI

5.4 Loop closure analysis for Sequence 00:

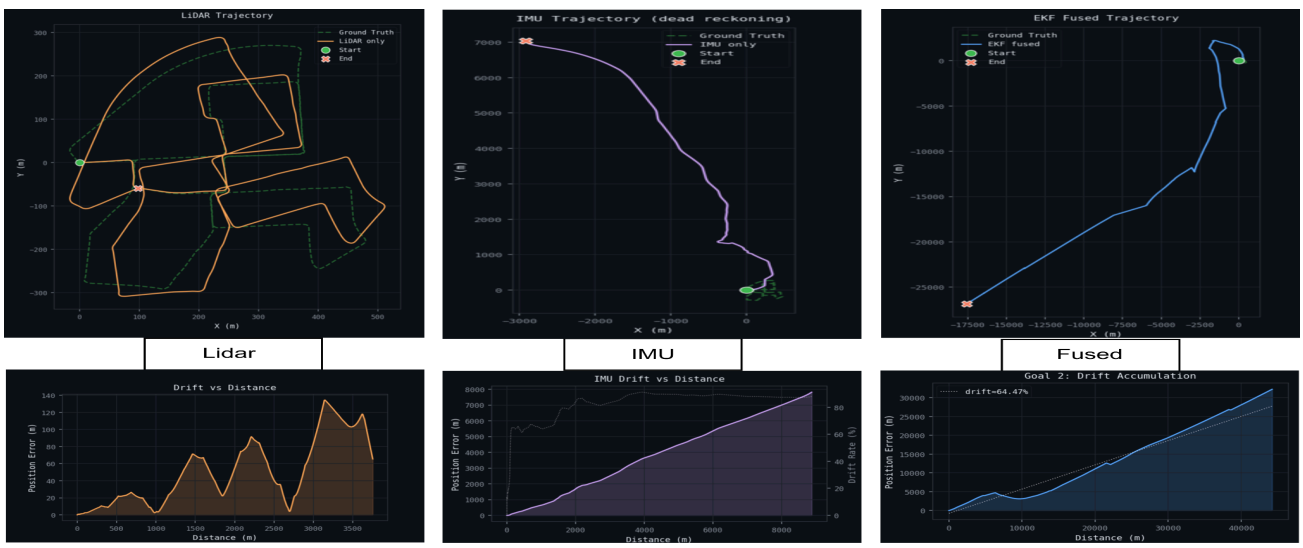


Figure 6 :Loop closure analysis for sequence 00, the plot contains plots of the trajectories obtained with Lidar-only, IMU-only and fused trajectory, and also compares the drift accumulated in each method with time)

5.5 Summary comparison table

The error here is the cumulative Trajectory Error(TE) in metres across the trajectory, the errors are not normalized here:

Driving Scenario	IMU-Only Error	LiDAR-Only Error	EKF Fusion Error
Urban, long, Loop closure(Sequence 00)	7815.44	65.40	32255.15
Rural, short, low speed (Sequence 04, Baseline)	164.84	8.28	183.23
Urban, medium-length, high speed (Sequence 01)	895.47	405.76	1701.16

Table 2: Cumulative Error for evaluated configurations for sequences(00, 01,04) of KITTI

6. Discussion

6.1 Trajectory Comparison and Errors

- Sequence 00, a 3.7 km closed urban loop, exposes the limitations of local-only fusion over long trajectories. IMU drift grows unbounded with time and distance,causing the errors in fused trajectories to blow out of proportion as seen in figure 2. The fused trajectory performs even worse beyond ~200 m (~20 s), as Lidar drift is incorporated rather than corrected, amplifying total error. This shows that local sensor fusion alone cannot maintain reliable pose estimation over large-scale trajectories without corrections of both sensors at loop closures.
- For Sequence 01 (high-speed highway, 72–90 km/h), motion distortion in Lidar scans combined with sparse environmental features leads to poor ICP convergence under limited iterations. Consequently, the fused trajectory error grows unbounded as unreliable Lidar updates fail to correct IMU drift, highlighting the limitations of Lidar–inertial fusion in feature-deprived, high-dynamic scenarios without motion-distortion compensation
- Sequence 04: As the shortest sequence with low structural complexity(recorded in feature-rich scenes), Sequence 04 shows fusion working effectively over short distances, reducing drift growth compared to IMU-only estimation. Lidar registration remains reliable enough to partially suppress IMU drift, but error still accumulates over time, confirming that fusion alone cannot ensure long-term consistency without global correction/ loop closure correction.

6.2 Loop Closure Analysis for Sequence 00

- As shown in the figure 6, Lidar pose error is corrected during loop closure, whereas IMU drift continues to accumulate.
- This mismatch leads the fusion system to combine corrected Lidar poses with drifted IMU estimates, resulting in inconsistent and inaccurate fused poses.
- A potential solution is to incorporate IMU bias correction or state reset after loop closure detection to maintain accurate pose estimation over longer trajectories

6.3 Failure Modes and Limitations

Several failure modes and limitations were identified:

- **Motion distortion of Point Clouds**: Lidar scans acquired over 100 ms contain points from slightly different vehicle poses. Without explicit deskewing (using IMU to correct individual point positions), ICP residuals are elevated, particularly at high vehicle speeds. Deskewing was not implemented in the current pipeline and represents the most significant potential improvement..
- **Voxel size tuning**: A fixed voxel size of 0.5 m is chosen to make the pipeline less computationally expensive but a smaller size is preferred for lower ICP RMSE values.
- **Reduced reliance on Lidar (higher-confidence sensor weighting):** In the current fusion setup, the state estimation is observed to be driven primarily by IMU-based prediction, with comparatively limited correction from Lidar measurements. This is reflected in the lower Kalman gain assigned to the Lidar update due to measurement noise tuning. As a result, while the fused pose is more accurate than IMU-only estimation, it still retains residual error and does not fully correct the drift.

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6.4 Operational Lifetime Without Global Correction

A key project goal was to assess how long the system can operate reliably without global corrections (GPS or loop closure). Based on the results for Sequence 04 as it was the baseline and other sequences were used to analyse different limitations:

- **IMU-only: ~5s**
- **Lidar-only: ~25s (for the entire dataset, as the ATE < 10m)**
- **EKF fusion: <10s, after which the ATE >25m**

7. Conclusion

This study showed that local Lidar-IMU fusion can improve pose estimation over short, well-conditioned trajectories, but its reliability degrades in long-duration, high-speed, or feature-deprived scenes . While Lidar can partially correct drift through geometric registration and loop closure, unbounded IMU drift remains a critical limitation when not explicitly corrected. In some cases, fusion can even amplify error when combining corrected Lidar poses with drifted inertial estimates. These results highlight that local sensor fusion alone is insufficient for consistent long-term localization without global correction mechanisms and uncertainty-aware estimation.

The future scope is:

- **IMU drift correction/reset:** Develop methods to correct the IMU drift, particularly when loop closures or trusted global references are detected.
- **Jacobian-based correction in the active window:** Apply Jacobian corrections within the current optimization or sliding window to better account for accumulated linearization and state estimation errors.
- **Uncertainty-aware fusion with covariance propagation:** Incorporate sensor measurement uncertainty/information matrices directly into the fusion process, while propagating covariances to improve the robustness of the pipeline and to reject outliers/ deteriorated sensor measurements.

8. References

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